

	Symbol	Value	Unit
Pressure	P		200 bar
Critical Pressure	Pc	45.99	bar
Reduced Pressure	Pr	4.348771	
gas constant	R	0.08314	L*bar/mol/K
Critical Temperature	Tc	190.6	K

s

Van der Waals Constants			Temperature (K)	Reduced Temp
omega	Ω	0.125	300	1.573976915
psi	Ψ	0.421875	305	1.600209864
b	b	0.04307	310	1.626442812
a	a	2.303489	315	1.652675761
			320	1.678908709
			325	1.705141658
			330	1.731374607
			335	1.757607555
			340	1.783840504
			345	1.810073452
			350	1.836306401
			355	1.862539349
			360	1.888772298
			365	1.915005247
			370	1.941238195
			375	1.967471144
			380	1.993704092
			385	2.019937041
			390	2.046169999
			395	2.072402938
			400	2.098635887
			405	2.124868835
			410	2.151101784
			415	2.177334732
			420	2.203567681
			425	2.22980063
			430	2.256033578
			435	2.282266527
			440	2.308499475
			445	2.334732424
			450	2.360965373
			455	2.387198321
			460	2.413434127
			465	2.439664218
			470	2.465897167
			475	2.492130115
			480	2.518363064
			485	2.544596013
			490	2.570828961
			495	2.597061191
			500	2.623294858
			505	2.649527807
			510	2.675760756
			515	2.701993704
			520	2.728226653
			525	2.754459601
			530	2.78069255
			535	2.806925498
			540	2.833158447
			545	2.859391396

Redlich-Kwong Constants			Temperature (K)	Reduced Temp
omega	Ω	0.08664	300	1.573976915
psi	Ψ	0.42748	305	1.600209864
beta	β	0.029853	310	1.626442812
a	a	2.334093	315	1.652675761
eccentric factor	w	0.012	320	1.678908709
m	m	0.503678	325	1.705141658

Soave-Redlich-Kwong Constants			Temperature (K)	Reduced Temp
omega	Ω	0.08664	300	1.573976915
psi	Ψ	0.42748	305	1.600209864
beta	β	0.029853	310	1.626442812
a	a	2.334093	315	1.652675761
eccentric factor	w	0.012	320	1.678908709
m	m	0.503678	325	1.705141658

Peng-Robinson Constants			Temperature (K)	Reduced Temp
omega	Ω	0.0778	300	1.573976915
psi	Ψ	0.45724	305	1.600209864
beta	β	0.026807	310	1.626442812
a	a	2.496587	315	1.652675761
K	K	0.393108	320	1.678908709
eccentric factor	w	0.012	325	1.705141658

Calculating Molar Volume												
Ideal Gas Law Value	Van der Waals		Redlich-Kwong			SRK		Peng-Robinson				
	Equation	Value	a(T)	Equation	Value	alpha(T)	Equation	Value	alpha(T)	Equation	Value	
0.12471	-1.0516E-12	0.102659	1.860454	3.42344E-06	0.105237	0.759987	4.31182E-06	0.10801	0.809859	-8.45372E-06	0.101208	
0.1267885	4.36224E-06	0.105201	1.845142	-4.06288E-07	0.108264	0.750871	-7.26079E-06	0.111193	0.802509	-5.94501E-06	0.104315	
0.128867	-5.50801E-06	0.10778	1.830201	5.37141E-06	0.111274	0.741884	1.03957E-05	0.114354	0.795252	-4.24486E-06	0.107404	
0.1309455	4.75486E-06	0.110389	1.815618	-1.41652E-05	0.114265	0.733023	-1.12142E-05	0.117491	0.788086	-3.0655E-06	0.110473	
0.133024	-2.47187E-05	0.113024	1.801377	-2.20637E-06	0.117235	0.724284	1.44217E-05	0.120603	0.781009	-2.23127E-06	0.111352	
0.1351025	3.0547E-05	0.11568	1.787467	3.31903E-06	0.120184	0.715665	1.04257E-05	0.123688	0.774019	1.02413E-05	0.116544	
0.137181	1.78022E-05	0.118351	1.773874	-3.57418E-07	0.123111	0.707163	4.65998E-06	0.126746	0.767113	7.72684E-06	0.119545	
0.1392595	5.08377E-06	0.121034	1.760586	2.85782E-06	0.126015	0.698775	-8.5288E-06	0.129777	0.760289	5.79615E-06	0.122524	
0.141338	0	0.123725	1.747593	-1.65477E-05	0.128897	0.6905	-2.57701E-06	0.132782	0.753547	4.29833E-06	0.125479	
0.1434165	-2.37135E-06	0.126421	1.734883	-7.18016E-06	0.131756	0.682333	2.78434E-06	0.13576	0.746884	3.1264E-06	0.128411	
0.145495	5.49281E-06	0.12912	1.722446	-2.1519E-06	0.134593	0.674274	1.23935E-06	0.138712	0.740298	2.20295E-06	0.13132	
0.1475735	-7.14294E-06	0.131819	1.710273	1.0023E-05	0.137408	0.66632	1.3186E-06	0.141639	0.733788	-1.09009E-05	0.134208	
0.149652	-3.18737E-07	0.134516	1.698355	1.0043E-05	0.140202	0.658468	7.20649E-07	0.144541	0.727352	-6.74389E-06	0.137074	
0.1517305	3.86914E-06	0.13721	1.686682	-1.2702E-07	0.142976	0.650716	-4.16896E-06	0.14742	0.720989	-3.29331E-06	0.139918	
0.153809	-2.05115E-07	0.139898	1.675247	-4.30512E-07	0.145729	0.643063	-1.67204E-06	0.150275	0.714696	3.34698E-06	0.142743	
0.1558875	-1.89667E-07	0.142581	1.664041	1.2295E-06	0.148463	0.635506	-5.21831E-07	0.153108	0.708474	-1.90269E-06	0.145547	
0.157966	-1.39694E-05	0.145257	1.653057	8.59126E-07	0.151178	0.628043	-1.14037E-07	0.155919	0.702319	-3.72169E-06	0.148332	
0.1600445	-6.22931E-06	0.147925	1.642288	-5.93092E-06	0.153875	0.620673	6.75826E-07	0.158709	0.696232	-5.1325E-06	0.151098	
0.162123	-2.24728E-06	0.150585	1.631727	-3.23146E-06	0.156554	0.613394	3.66968E-08	0.161478	0.69021	-6.21505E-06	0.153847	
0.1642015	0	0.153236	1.621366	-8.8044E-08	0.159216	0.606204	-3.04681E-10	0.164229	0.684253	-7.03277E-06	0.156577	
0.16628	-2.27273E-07	0.155879	1.611201	-6.15347E-07	0.161862	0.599101	-3.9536E-08	0.16696	0.678359	-7.63635E-06	0.159291	
0.1683585	-4.9444E-07	0.158512	1.601224	-2.02044E-07	0.164493	0.592084	-4.64971E-07	0.169673	0.672526	-8.06655E-06	0.161989	
0.170437	1.74129E-06	0.161136	1.591431	-1.07588E-07	0.167107	0.585151	-1.82441E-06	0.172369	0.666755	-8.35622E-06	0.164671	
0.1725155	2.97732E-06	0.163751	1.581815	4.1103E-07	0.169708	0.578301	-1.48953E-07	0.175048	0.661043	-8.53209E-06	0.167337	
0.174594	-2.56112E-06	0.166356	1.572371	3.6464E-08	0.172294	0.571532	-3.46537E-07	0.17771	0.65539	-8.61592E-06	0.169989	
0.1766725	1.40576E-06	0.168951	1.563094	-2.6622E-08	0.174866	0.564842	-6.59282E-07	0.180357	0.649795	-8.62547E-06	0.172627	
0.178751	1.41434E-06	0.171537	1.55398	-4.4885E-09	0.177426	0.558231	-1.09017E-06	0.182989	0.644256	-8.57535E-06	0.175251	
0.1808295	-6.53858E-08	0.174114	1.545023	-1.05966E-07	0.179973	0.551697	-1.62434E-06	0.185606	0.638773	-8.47756E-06	0.177861	
0.182908	7.33004E-07	0.176682	1.53622	-5.30555E-07	0.182507	0.545238	-2.23107E-06	0.188209	0.633345	-8.34203E-06	0.180459	
0.1849865	-5.77346E-06	0.179241	1.527565	-8.53841E-08	0.18503	0.538854	-2.86735E-06	0.190798	0.62797	-8.17687E-06	0.183044	
0.187065	-2.19768E-07	0.18179	1.519055	-9.7688E-08	0.187542	0.532542	-3.48225E-06	0.193375	0.622648	-7.98884E-06	0.185618	
0.1891435	-1.3005E-07	0.184331	1.510685	-2.0508E-07	0.190042	0.526303	-4.02111E-06	0.195938	0.617378	-7.78351E-06	0.188179	
0.191222	-1.21044E-06	0.186863	1.502453	-3.69743E-07	0.192532	0.520134	-4.42931E-06	0.198489	0.612159	-7.56549E-06	0.19073	
0.1933005	-6.15148E-07	0.189387	1.494353	-5.97018E-07	0.195012	0.514034	-4.6553E-06	0.201029	0.60699	-7.33857E-06	0.19327	
0.195379	-2.82342E-07	0.191902	1.486383	-8.86485E-07	0.197482	0.508003	-4.65301E-06	0.203557	0.601871	-7.10587E-06	0.195799	
0.1974575	-1.13599E-07	0.19441	1.478539	-6.85538E-08	0.199943	0.50204	-4.38333E-06	0.206074	0.5968	-6.87001E-06	0.198318	
0.199536	-2.39128E-07	0.196909	1.470818	-1.62248E-06	0.202395	0.496142	-3.81513E-06	0.20858	0.591778	-6.63309E-06	0.200827	
0.2016145	-9.8938E-08	0.199401	1.463217	-1.13294E-07	0.204837	0.49031	-2.92562E-06	0.211076	0.586802	-6.3968E-06	0.203326	
0.203693	1.42795E-06	0.201885	1.455733	-1.37008E-07	0.207271	0.484542	-1.70029E-06	0.213562	0.581873	-6.16257E-06	0.205817	
0.2057715	5.09688E-07	0.204362	1.448362	-1.60148E-07	0.209697	0.478837	-1.28315E-06	0.216038	0.57699	-5.93153E-06	0.208298	
0.20785	1.30466E-07	0.206831	1.441102	-3.28638E-06	0.212115	0.473195	-1.69049E-06	0.218505	0.572152	-5.70458E-06	0.210771	
0.2099285	1.71516E-08	0.209293	1.43395	-2E-07	0.214526	0.467613	-3.63337E-06	0.220963	0.567358	-5.48245E-06	0.213235	
0.212007	-9.95275E-08	0.211749	1.426903	-3.89093E-06	0.216928	0.462093	-5.66978E-06	0.223412	0.562607	-5.26564E-06	0.215691	
0.2140855	-4.91276E-08	0.214198	1.41996	-2.23689E-07	0.219324	0.456631	-7.77291E-06	0.225852	0.5579	-5.05462E-06	0.218139	
0.216164	-1.01792E-08	0.21664	1.413117	-4.12702E-06	0.221712	0.451229	-9.91805E-06	0.228285	0.553235	-4.84965E-06	0.220579	
0.2182425	-7.34025E-08	0.219076	1.406371	-2.2319E-07	0.224094	0.445884	-1.20829E-05	0.230709	0.548612	-4.65097E-06	0.223012	
0.220321	-2.54371E-07	0.221506	1.399722	-2.1185E-07	0.226469	0.440596	-1.68313E-06	0.233125	0.54403	-4.45869E-06	0.225438	
0.2223995	-6.20164E-07	0.223929	1.393166	-1.92293E-07	0.228838	0.435365	-1.9639E-06	0.235534	0.539489	-4.27293E-06	0.227856	
0.224478	-1.22915E-06	0.226347	1.386701	-1.64071E-07	0.231201	0.430189	-2.24604E-06	0.237935	0.534988	-4.09366E-06	0.230268	
0.2265565	-3.97489E-07	0.228759	1.380325	-1.26861E-07	0.233557	0.425067	-2.2573E-06	0.24033	0.530527	-3.92089E-06	0.232673	

550	2.885624344	0.228635	-8.30175E-07	0.231165	1.374037	-8.04651E-08	0.235908	0.42	-2.80572E-06	0.242717	0.526104	-3.75451E-06	0.235072
555	2.911857293	0.2307135	-1.49033E-06	0.233566	1.367833	-2.46589E-07	0.238254	0.414985	-3.07957E-06	0.245098	0.52172	-3.59445E-06	0.237464
560	2.938090241	0.232792	-2.40751E-06	0.235961	1.361713	-3.81214E-07	0.240593	0.410023	-3.34735E-06	0.247472	0.517374	-3.44061E-06	0.23985
565	2.96432319	0.2348705	-1.11948E-07	0.238351	1.355675	-1.06876E-07	0.242928	0.405113	-3.60783E-06	0.24984	0.513065	-3.29284E-06	0.24223
570	2.990556139	0.236949	-1.73622E-07	0.240736	1.349716	-3.26135E-06	0.245257	0.400254	-3.85994E-06	0.252201	0.508793	-3.15098E-06	0.244604
575	3.016789087	0.2390275	-2.53868E-07	0.243117	1.343834	-2.50944E-07	0.247582	0.395445	-4.10286E-06	0.254557	0.504558	-3.0149E-06	0.246973
580	3.043022036	0.241106	-3.53882E-07	0.245492	1.338029	-3.26096E-07	0.249901	0.390686	-4.3359E-06	0.256907	0.500358	-2.88441E-06	0.249336
585	3.069254984	0.2431845	-4.74177E-07	0.247862	1.332299	-7.40619E-06	0.252216	0.385976	-4.55856E-06	0.259251	0.496194	-2.75934E-06	0.251694
590	3.095487933	0.245263	-6.14578E-07	0.250228	1.326642	-8.83755E-06	0.254527	0.381315	-4.77049E-06	0.261589	0.492065	-2.63951E-06	0.254046
595	3.121720881	0.2473415	-7.74257E-07	0.25259	1.321056	-1.02801E-05	0.256832	0.376701	-4.97145E-06	0.263922	0.487971	-2.52476E-06	0.256394
600	3.14795383	0.24942	-9.51785E-07	0.254947	1.31554	-1.17267E-05	0.259134	0.372135	-5.16131E-06	0.26625	0.483911	-2.41491E-06	0.258736
605	3.174186779	0.2514985	-6.32941E-08	0.257299	1.310093	-8.29997E-08	0.261431	0.367615	-5.34003E-06	0.268573	0.479884	-2.30976E-06	0.261074
610	3.200419727	0.253577	-7.46494E-08	0.259648	1.304712	-9.29584E-08	0.263724	0.363142	-5.50768E-06	0.270891	0.475891	-2.20915E-06	0.263406
615	3.226652676	0.2556555	-8.6609E-07	0.261992	1.299398	-1.02965E-07	0.266014	0.358713	-5.66438E-06	0.273204	0.471931	-2.11288E-06	0.265735
620	3.252885624	0.257734	-1.0362E-06	0.264332	1.294148	-1.12979E-07	0.268299	0.35433	-5.81027E-06	0.275513	0.468003	-2.02081E-06	0.268058
625	3.279118573	0.2598125	-1.21842E-06	0.266669	1.288891	-1.22962E-07	0.270581	0.349992	-5.9456E-06	0.277816	0.464108	-1.93278E-06	0.270378
630	3.305351522	0.261891	-1.4107E-06	0.269001	1.283836	-2.45761E-06	0.272858	0.345697	-6.07063E-06	0.280115	0.460244	-1.84859E-06	0.272693
635	3.33158447	0.2639695	-1.61075E-06	0.27133	1.278771	-1.427E-07	0.275133	0.341445	-6.18562E-06	0.28241	0.456412	-1.76811E-06	0.275004
640	3.357817419	0.266048	-1.81613E-06	0.273655	1.273766	-1.52396E-07	0.277404	0.337237	-6.2909E-06	0.284701	0.45261	-1.69119E-06	0.277311
645	3.384050367	0.2681265	-2.0243E-06	0.275977	1.268819	-1.61945E-07	0.279671	0.33307	-6.38677E-06	0.286987	0.448839	-1.61765E-06	0.279614
650	3.410283316	0.270205	-1.42436E-06	0.278295	1.263393	-1.71324E-07	0.281935	0.328946	-6.47358E-06	0.28927	0.445099	-1.54737E-06	0.281913
655	3.436516264	0.2722835	-2.43855E-06	0.280609	1.259097	-1.80514E-07	0.284196	0.324863	-6.55164E-06	0.291548	0.441388	-1.48021E-06	0.284208
660	3.462749213	0.274362	-2.63944E-06	0.282921	1.254318	-3.51556E-06	0.286454	0.320821	-6.62133E-06	0.293823	0.437707	-1.41602E-06	0.286499
665	3.488982162	0.2764405	-1.54896E-07	0.285229	1.249594	-1.98271E-07	0.288708	0.316819	-6.68296E-06	0.296094	0.434055	-1.35469E-06	0.288787
670	3.51521511	0.278519	-1.1852E-07	0.287534	1.244922	-3.84028E-06	0.29096	0.312858	-6.73687E-06	0.298361	0.430432	-1.29608E-06	0.291071
675	3.541448059	0.2805975	-1.28608E-07	0.289835	1.240303	-3.99628E-06	0.293209	0.308935	-6.78339E-06	0.300624	0.426838	-1.24009E-06	0.293352
680	3.567681007	0.282676	-2.53775E-06	0.292134	1.235735	-4.14784E-06	0.295455	0.305052	-6.82285E-06	0.302884	0.423272	-1.18657E-06	0.29563
685	3.593913956	0.2847545	-1.4801E-07	0.294429	1.231216	-2.30985E-07	0.297698	0.301208	-6.85555E-06	0.305141	0.419734	-1.13545E-06	0.297904
690	3.620146905	0.286833	-1.57122E-07	0.296722	1.226747	-2.38542E-07	0.299938	0.297402	-6.88181E-06	0.307394	0.416223	-1.0866E-06	0.300175
695	3.646379853	0.2889115	-1.65716E-07	0.299012	1.222327	-4.575E-06	0.302176	0.293634	-6.90191E-06	0.309644	0.41274	-1.03992E-06	0.302442
700	3.672612802	0.29099	-3.19153E-06	0.301299	1.217953	-2.52887E-07	0.304411	0.289903	-6.91615E-06	0.31189	0.409284	-9.95313E-07	0.304707
705	3.69884575	0.2930685	-3.32764E-06	0.303583	1.213627	-2.59675E-07	0.306644	0.286209	-6.92479E-06	0.314134	0.405855	-9.5269E-07	0.306969
710	3.725078699	0.295147	-3.44982E-06	0.305865	1.209346	-2.66207E-07	0.308874	0.282552	-6.92812E-06	0.316375	0.402452	-9.11961E-07	0.309227
715	3.751311647	0.2972255	-1.93247E-07	0.308143	1.20511	-2.72485E-07	0.311102	0.278931	-6.92636E-06	0.318612	0.399076	-8.73042E-07	0.311483
720	3.777544596	0.299304	-1.98061E-07	0.31042	1.200918	-2.78513E-07	0.313327	0.275346	-6.91977E-06	0.320847	0.395725	-8.35849E-07	0.313736
725	3.803777545	0.3013825	-2.0193E-07	0.312693	1.19677	-2.84293E-07	0.31555	0.271796	-6.90859E-06	0.323078	0.3924	-8.00299E-07	0.315986
730	3.830010493	0.303461	-3.27177E-06	0.314965	1.192664	-2.89829E-07	0.317771	0.268282	-6.89304E-06	0.325307	0.3891	-7.66326E-07	0.318233
735	3.856243442	0.3055395	-1.8337E-07	0.317233	1.188601	-5.50925E-06	0.31999	0.264802	-6.87335E-06	0.327534	0.385826	-7.33856E-07	0.320478
740	3.88247639	0.307618	-1.88602E-07	0.3195	1.184578	-3.00189E-07	0.322206	0.261357	-6.8497E-06	0.329757	0.382577	-7.02822E-07	0.32272
745	3.908709339	0.3096965	-1.93126E-07	0.321764	1.180597	-3.05023E-07	0.324421	0.257946	-6.8223E-06	0.331978	0.379352	-6.73156E-07	0.324959
750	3.934942288	0.311775	-1.96906E-07	0.324025	1.176655	-3.09632E-07	0.326633	0.254568	-6.79134E-06	0.334196	0.376151	-6.44794E-07	0.327196
755	3.961175236	0.3138535	-1.99909E-07	0.326285	1.172752	-3.14023E-07	0.328843	0.251224	-6.757E-06	0.336412	0.372975	-6.17684E-07	0.329431
760	3.987408185	0.315932	-3.73684E-06	0.328542	1.168888	-3.18201E-07	0.331051	0.247913	-6.71944E-06	0.338625	0.369822	-5.9176E-07	0.331663
765	4.013641133	0.3180105	-2.0348E-07	0.330797	1.165062	-3.22172E-07	0.333258	0.244635	-6.67886E-06	0.340836	0.366694	-5.66976E-07	0.333893
770	4.039874082	0.320089	-2.04007E-07	0.333305	1.161273	-3.25943E-07	0.335462	0.241389	-6.63539E-06	0.343045	0.363588	-5.43279E-07	0.33612
775	4.06610703	0.3221675	-3.77086E-06	0.335301	1.157521	-6.16818E-06	0.337665	0.238175	-6.58921E-06	0.345251	0.360506	-5.20615E-07	0.338345
780	4.092339979	0.324246	-2.02477E-07	0.337549	1.153805	-3.32904E-07	0.339866	0.234993	-6.54044E-06	0.347455	0.357447	-4.98941E-07	0.340568
785	4.118572928	0.3263245	-2.00404E-07	0.339796	1.150124	-3.36108E-07	0.342065	0.231843	-6.48926E-06	0.349657	0.35441	-4.78214E-07	0.342789
790	4.144805876	0.328403	-3.66027E-06	0.342041	1.146479	-3.39134E-07	0.344262	0.228723	-6.43577E-06	0.351856	0.351397	-4.58388E-07	0.345008
795	4.171038415	0.3304815	-1.93632E-07	0.344284	1.142868	-3.41988E-07	0.346458	0.225635	-6.38012E-06	0.354054	0.348405	-4.39422E-07	0.347224
800	4.197271773	0.33256	-2.00691E-07	0.346525	1.139291	-3.44677E-07	0.348652	0.222577	-6.32242E-06	0.356249	0.345436	-4.21282E-07	0.349439
805	4.223504722	0.3346385	-2.00676E-07	0.348764	1.135747	-3.47206E-07	0.350844	0.219549	-6.26283E-06	0.358442	0.342488	-4.03925E-07	0.351651
810	4.249737671	0.336717	-3.71308E-06	0.351001	1.132237	-3.4958E-07	0.353035	0.216552	-6.20143E-06	0.360634	0.339562	-3.87315E-07	0.353862
815	4.275970619	0.3387955	-1.98634E-07	0.353236	1.128758	-3.51805E-07	0.355224	0.213584	-6.13836E-06	0.362823	0.336658	-3.71426E-07	0.35607
820	4.302203568	0.340874	-3.65299E-06	0.35547	1.125311	-3.53886E-07	0.357412	0.210646	-6.07372E-06	0.36501	0.333775	-3.56219E-07	0.358277
825	4.328436516	0.3429525	-3.60425E-06	0.357701	1.121896	-3.55829E-07	0.359598	0.207736	-6.00761E-06	0.367196	0.330913	-3.41664E-07	0.360481
830	4.354669465	0.345031	-1.90559E-07	0.359932	1.118512	-3.57638E-07	0.361782	0.204856	-5.94014E-06	0.36938	0.328072	-3.27735E-07	0.362684
835	4.380902413	0.3471095	-1.86549E-07	0.36216	1.115158	-3.59319E-07	0.363966	0.202005	-5.87141E-06	0.371561	0.325252	-3.14401E-07	0.364885
840	4.407135362	0.349188	-1.81894E-07	0.364387	1.111834	-6.78145E-06	0.366147	0.199182	-5.80152E-06	0.373741	0.322453	-3.01636E-07	0.367085
845	4.433368311	0.3512665	-3.28707E-06	0.366612	1.10854	-3.62312E-07	0.368328	0.196387	-5.73056E-06	0.37592	0.319673	-2.89415E-07	0.369282

850	4.459601259	0.353345	-1.70691E-07	0.368835	1.105275	-3.63633E-07	0.370507	0.19362	-5.65861E-06	0.378096	0.316914	-2.77716E-07	0.371478
855	4.485834208	0.3554235	-1.64168E-07	0.371057	1.102038	-3.64845E-07	0.372684	0.19088	-5.58577E-06	0.380271	0.314175	-2.66513E-07	0.373672
860	4.512067156	0.357502	-1.57051E-07	0.373278	1.09883	-3.65949E-07	0.374861	0.188169	-5.5121E-06	0.382444	0.311456	-2.55785E-07	0.375865
865	4.538300105	0.3595805	-1.49354E-07	0.375497	1.095649	-6.90491E-06	0.377036	0.185484	6.53836E-07	0.384616	0.308756	-2.45507E-07	0.378055
870	4.564533054	0.361659	-2.63042E-06	0.377714	1.092496	-3.67855E-07	0.37921	0.182826	-5.36268E-06	0.386786	0.306076	-2.35668E-07	0.380245
875	4.590766002	0.3637375	-2.46704E-06	0.37993	1.089371	-6.94072E-06	0.381382	0.180195	-5.28707E-06	0.388955	0.303416	-2.26241E-07	0.382432
880	4.616998951	0.365816	-2.29372E-06	0.382144	1.086271	-6.956E-06	0.383554	0.17759	-5.21094E-06	0.391122	0.300774	-2.1721E-07	0.384619
885	4.643231899	0.3678945	-2.11079E-06	0.384357	1.083198	-6.96961E-06	0.385724	0.175012	-5.13438E-06	0.393287	0.298151	-2.08558E-07	0.386803
890	4.669464848	0.369973	-1.9186E-06	0.386569	1.080151	-6.98163E-06	0.387893	0.17246	-5.05746E-06	0.395451	0.295548	-2.00269E-07	0.388986
895	4.695697796	0.3720515	-1.71752E-06	0.388779	1.07713	-3.7102E-07	0.39006	0.169934	-4.98024E-06	0.397613	0.292962	-1.92325E-07	0.391168
900	4.721930745	0.37413	-8.07384E-08	0.390988	1.074134	-3.71405E-07	0.392227	0.167433	-4.90278E-06	0.399774	0.290396	-1.84712E-07	0.393348
905	4.748163694	0.3762085	-6.90583E-08	0.393196	1.071163	-3.71716E-07	0.394393	0.164957	-4.82514E-06	0.401934	0.287848	-1.77416E-07	0.395527
910	4.774396642	0.378287	-5.69683E-08	0.395402	1.068216	-3.71955E-07	0.396557	0.162507	-4.74739E-06	0.404092	0.285318	-1.70423E-07	0.397705
915	4.800629591	0.3803655	-4.55559E-07	0.397607	1.065293	-3.72123E-07	0.398721	0.160082	-4.66957E-06	0.406249	0.282806	-1.63721E-07	0.399881
920	4.826862539	0.382444	-3.22386E-07	0.399811	1.062394	-3.72226E-07	0.400883	0.157681	-4.59175E-06	0.408405	0.280312	-1.57295E-07	0.402056
925	4.853095488	0.3845225	-1.87006E-07	0.402013	1.059519	-3.72264E-07	0.403044	0.155306	-4.51397E-06	0.410559	0.277835	-1.51133E-07	0.404229
930	4.879328437	0.386601	-5.00658E-07	0.404214	1.056667	-7.02678E-06	0.405205	0.152954	-4.43629E-06	0.412712	0.275377	-1.45225E-07	0.406401
935	4.905561385	0.3886795	-1.66488E-06	0.406414	1.053838	-3.72156E-07	0.407364	0.150627	-4.35874E-06	0.414864	0.272936	-1.39561E-07	0.408572
940	4.931794334	0.390758	-2.26587E-07	0.408613	1.051032	-3.72016E-07	0.409522	0.148323	-4.28139E-06	0.417015	0.270512	-1.34129E-07	0.410742
945	4.958027282	0.3928365	-3.63025E-07	0.41081	1.048247	-3.7182E-07	0.41168	0.146044	-4.20427E-06	0.419164	0.268105	-1.28917E-07	0.41291
950	4.984260231	0.394915	-4.98208E-07	0.413007	1.045485	-7.02051E-06	0.413836	0.143788	-4.12742E-06	0.421312	0.265716	-1.2392E-07	0.415078
955	5.010493179	0.3969935	-6.39808E-08	0.415202	1.042745	-3.71273E-07	0.415992	0.141555	-4.05089E-06	0.423459	0.263343	-1.19125E-07	0.417244
960	5.036726128	0.399072	-7.76944E-08	0.417396	1.040026	-3.70925E-07	0.418146	0.139346	-3.97471E-06	0.425605	0.260987	-1.14525E-07	0.419408
965	5.062959077	0.4011505	-9.13569E-08	0.419589	1.037328	-3.7053E-07	0.4203	0.13716	-3.89894E-06	0.42775	0.258648	-1.10113E-07	0.421572
970	5.089192025	0.403229	-1.04958E-07	0.421781	1.034651	-3.70089E-07	0.422453	0.134997	-3.82357E-06	0.429893	0.256326	-1.05879E-07	0.423735
975	5.115424974	0.4053075	-1.18487E-07	0.423972	1.031994	-3.69605E-07	0.424605	0.132856	-3.74868E-06	0.432036	0.254019	-1.01816E-07	0.425896
980	5.141657922	0.407386	-1.31934E-07	0.426162	1.029358	-3.6908E-07	0.426756	0.130738	-3.67425E-06	0.434177	0.251729	-9.79158E-08	0.428056
985	5.167890871	0.4094645	-1.45292E-07	0.428351	1.026743	-3.68514E-07	0.428906	0.128642	-3.60036E-06	0.436317	0.249456	-9.41732E-08	0.430216
990	5.19412382	0.411543	-1.58552E-07	0.430538	1.024146	-3.67909E-07	0.431055	0.126568	-3.52702E-06	0.438457	0.247198	-9.05818E-08	0.432374
995	5.220356768	0.4136215	-1.71707E-07	0.432725	1.02157	-3.67268E-07	0.433204	0.124517	-3.45426E-06	0.440595	0.244956	-8.71327E-08	0.434531
1000	5.246589717	0.4157	-3.46813E-06	0.434911	1.019013	-6.94111E-06	0.435352	0.122487	-3.38209E-06	0.442732	0.24273	2.84564E-06	0.436687